Learning Objective 2.4: I can describe how reflectors, Yagi-Uda antennas, and arrays achieve high gain, and select an appropriate high-gain antenna for a given application.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Sizing a C-band earth station.** A ground station uses a **3.0 m diameter** prime-focus parabolic reflector at **6.0 GHz**. Take the aperture efficiency to be $\eta_{\text{ap}} = 0.65$ and $c = 3 \times 10^8$ m/s.

   (a) In one or two sentences, explain why the statement "gain is fundamentally a statement about area" is more useful to a designer than "gain is how much the antenna amplifies."

   (b) Find the wavelength, the dish diameter in wavelengths, and the gain in dBi.

   G = $\boxed{}$

   (c) Estimate the half-power beamwidth, and state in one sentence what it implies about pointing.

   θ_{HP} = $\boxed{}$

   (d) Management wants **3 dB more gain** at the same frequency. What dish diameter does that take, and name one specific cost of the change.

2. **Where the efficiency went.** Same 3.0 m dish. The manufacturer quotes the following loss factors: spillover 0.88, illumination taper 0.86, blockage 0.96, and a lumped "everything else" factor of 0.98. The reflector surface is held to **0.5 mm RMS**. Ruze's formula gives the surface penalty as loss (dB) $= 685.8\,(\sigma/\lambda)^2$.

   (a) Spillover and illumination taper trade directly against each other. Explain the trade in one or two sentences, and state the rule of thumb for where the optimum sits.

   (b) Compute the aperture efficiency from the four quoted factors, before any surface error. Call it $\varepsilon_{ap,1}$.

   $\varepsilon_{ap,1}$ =

   (c) Add the Ruze surface term at 6.0 GHz and report the total aperture efficiency.

   η_{ap} =

   (d) Someone proposes reusing this *same physical dish* at **30 GHz**. Recompute the surface term and the total aperture efficiency, then say in one or two sentences what the result means for the proposal.

3. **Reading a Yagi-Uda.** A Yagi-Uda has one driven element, one reflector behind it, and a row of directors in front along a boom.

   (a) The reflector is cut slightly *longer* than resonance and the directors slightly *shorter*. Explain what that detuning does electrically, and how it produces an endfire beam.

   (b) A second reflector buys almost nothing, but a seventh director still helps. Why?

   (c) A 6-element Yagi on a 1.0λ boom gives 10 dBi. Using the rule of thumb of $+3$ dB per doubling of boom length, estimate the boom needed for 16 dBi at 435 MHz. Then say whether you believe the answer.

   L =

   (d) Yagis are narrowband — a few percent. Explain why, and name one application where that does not matter.

4. **Choosing an antenna.** You need a **25 dBi** antenna at each end of a 2 km rooftop-to-rooftop link at **5.8 GHz**. The mounts are fixed (no steering), the site is windy, and the budget is small.

   (a) Find the required effective aperture, then the diameter of a parabolic dish that delivers it with $\eta_{\text{ap}} = 0.60$.

   D =

   (b) A vendor offers a flat patch array instead, with $\eta_{\text{ap}} = 0.75$ and elements on a $\lambda/2$ grid. How many elements, and how big is the panel?

   N =

   (c) Pick one and defend it in two or three sentences, using the numbers you just computed.

   (d) The dish beam is 9.6° wide. Using the beam-shape approximation $L(\theta) \approx 3\,(2\theta/\theta_{HP})^2$ dB, how far off boresight can the mount drift before you lose 0.5 dB?

   θ =

5. **Does the link close?** Continue the 5.8 GHz, 2 km link from the previous question. The transmitter delivers **20 dBm** and the receiver sensitivity is -80 **dBm**. Use Friis, $P_r = P_t G_t G_r \left(\lambda / 4\pi R\right)^2$.

   (a) Compute the free-space path loss in dB.

   L_{fs} = [ ]

   (b) Find the received power with the 25 dBi dishes at both ends, and the link margin.

   P_r = [ ]

   (c) Now replace both dishes with 8 dBi patches. Recompute P_r and the margin.

   P_r = [ ]

   (d) Which link would you actually deploy, and why? Name two specific things that eat margin on a real rooftop link.

Documentation: ______________________________